A Construction of Evolving k-threshold Secret Sharing Scheme over A Polynomial Ring

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Abstract. The threshold secret sharing scheme enables a dealer to distribute the share to every participant such that the secret is correctly recovered from a certain amount of shares. The traditional (k, n) threshold secret sharing scheme requires that the number of participants n is known in advance. In contrast, the evolving secret sharing scheme allows that n can be uncertain and even ever-growing. In this paper, we consider the evolving secret sharing scenario. Based on the prefix codes, we propose a brand-new construction of evolving k-threshold secret sharing scheme for an ℓ-bit secret over a polynomial ring, with correctness and perfect security. The proposed scheme is the first evolving k-threshold secret sharing scheme by generalizing Shamir's scheme onto a polynomial ring. Besides, the proposed scheme also establishes the connection between prefix codes and the evolving schemes for $k \geq 2$. The analysis shows that the size of the t-th share is $(k-1)(\ell_t - 1) + \ell$ bits, where ℓ_t denotes the length of a binary prefix code of encoding integer t. In particular, when δ code is chosen as the prefix code, the share size is $(k-1)\lfloor \lg t \rfloor + 2(k-1)\lfloor \lg(\lfloor \lg t \rfloor + 1)\rfloor + \ell$, which improves the prior best result $(k-1)\lg t + 6k^4 \ell \lg\lg t \cdot \lg\lg\lg t + 7k^4 \ell \lg k$, where lg denotes the binary logarithm. Specifically, when $k = 2$, the proposal also provides a unified mathematical decryption for prior evolving 2-threshold secret sharing schemes and also achieves the minimal share size for a single-bit secret, which is the same as the best-known scheme.

Keywords: Threshold secret sharing · Evolving · Prefix codes · Polynomial ring · Share size · Security.

1 Introduction

The (k, n) secret sharing scheme encodes the secret into n shares such that the secret can be losslessly recovered from any k out of n shares, and any $k-1$ shares cannot decode any information for the secret. Shamir [1] and Blakley [2] independently proposed the (k,n) secret sharing scheme in 1979. Due to the practicality of secret sharing scheme, it has been widely used and plays a very important role in modern cryptography [3–6]. However, the conventional secret sharing schemes require that the dealer knows the maximum number of participants n in advance. When we allow the dealer can produce the corresponding share for each new participant, the maximum number of n cannot be determined, and hence, most conventional schemes cannot be applied to this scenario.

To solve this issue, Komargodski et al. [7, 8] introduced an evolving k-threshold secret sharing scheme. In this scheme, the secret can be recovered from any k out of infinitely many participants, and any $k-1$ shares out of these cannot get any information about the secret. Specifically, Komargodski et al. [8] first proposed the construction of an evolving 2-threshold secret sharing scheme based on prefix codes, and showed the equivalence between binary prefix codes for the integers and evolving 2-threshold secret sharing schemes. The results show that when a set of prefix codes is given, there must exist a corresponding construction of the evolving 2-threshold scheme, and vice versa. Later, D'Arco et al. [9] also provided an alternative technique to show the equivalence. These results links prefix codes with evolving 2-threshold scheme, reducing the complex task of constructing such schemes to the simpler task of designing prefix codes, which holds significant research value. By analyzing the corresponding share size, it [8] shows that the share size of the t-th share is no more than $\lg t + (\ell+1)\lg\lg t + 4\ell + 1$ bits for an ℓ-bit secret. The scheme is optimal for a 1-bit secret. Furthermore, the authors in [8] proposed a general construction of the evolving k-threshold secret sharing scheme for arbitrary k, and analyzed that the t-th share size of the evolving k-threshold scheme is $(k-1)\lg t + 6k^4\ell \lg\lg t \cdot \lg\lg\lg t + 7k^4\ell\lg k$.

However, firstly, the evolving schemes in [8] are constructed based on the idea of distributing shares on a generational basis and use an evolving scheme once and Shamir's scheme multiple times. Hence, they will become increasingly complex and even difficult to construct as k increases. The natural idea for constructing the evolving scheme is based on simulating Shamir's scheme. Komargodski et al. [8] attempted this approach but failed. Thus, how to design algebraic-oriented constructions for evolving secret sharing schemes is also left as an open problem in [8]. Secondly, it is unknown whether there are connections between prefix codes and the evolving k-threshold secret sharing schemes when $k > 2$. This is also left as an open problem in [8]. Thirdly, the corresponding share size in [8] is not optimal. To further reduce the share size, D'Arco et al. [10] proposed a new construction of the evolving 3-threshold scheme based on the Chinese Remainder Theorem, which is slightly different from the model in [8]. They achieved the share size of the t-th participant close to $\lg t + poly(k)\cdot o(\lg t)$. However, the scheme can only be used in the case of $k=3$. As for $k \geq 4$, there are no related works, to the best of our knowledge.

To this end, we propose a brand-new construction of an evolving k-threshold secret sharing scheme for an ℓ-bit secret over a polynomial ring based on prefix codes, where the size of the t-th share is determined by the codeword length for encoding t. The proposed schemes achieve algebraic-oriented constructions by generalizing Shamir's scheme and establish the connection between prefix codes and the evolving schemes for $k \geq 2$. Specifically, we first propose the construction of an evolving 2-threshold secret sharing scheme over $\mathbb{F}_2[x]$. We also apply the construction to other complicated binary prefix codes to compare the share size with the scheme proposed in [8]. It shows that the proposed scheme can achieve a smaller size for some binary prefix codes. Second, we extend the scheme to the evolving k-threshold secret sharing scheme on $\mathbb{F}_2[x]$ for $k \geq 3$. The analysis of share size shows that the t-th share of the proposed scheme is $(k-1)(\ell_t - 1) + \ell$ bits, where ℓ_t denotes the length of a prefix code of encoding integer t. When using δ code [11] as the prefix code, the share size is given by $(k-1)\lfloor \lg t \rfloor + 2(k-1)\lfloor \lg(\lfloor \lg t \rfloor + 1) \rfloor + \ell$, which is smaller than the prior result $(k-1)\lg t + 6k^4 \ell \lg\lg t \cdot \lg\lg\lg t + 7k^4 \ell \lg k$ in [8]. Third, considering the secret $s \in \{0, 1, \cdots, p-1\}^\ell$, we proposed a construction of evolving k-threshold scheme over the polynomial ring $\mathbb{F}_p[x]$, where $k \geq 2$. This can be seen as an extension of the proposed scheme. In addition, we also show some p-ary prefix codes for positive integers and then apply the construction to these codes.

1.1 Our Contributions

Consequently, the above interesting and challenging problems remain open. In this paper, we focus on settling these problems simultaneously. Then the major contributions of this paper are enumerated as follows.

- The proposed brand-new scheme is more direct and concise by using the algebraic structure of polynomial ring. This result addresses the open question raised by [8]. It is the first evolving k-threshold secret sharing scheme that generalizes Shamir's scheme to a polynomial ring. Given that Shamir's scheme is the standard in traditional secret sharing, we believe that the proposed scheme has the potential to become the benchmark for evolving schemes.
- The proposed scheme successfully establishes the connection between prefix codes and evolving k-threshold secret sharing for arbitrary k, where the t-th share size is determined by the codeword length of encoding t. This demonstrates that there is a corresponding construction of an evolving k-threshold secret sharing scheme for any given prefix code. This result can answer another open question raised by [8].
- The share size of the proposed evolving k-threshold schemes is $(k-1)(\ell_t - 1) + \ell$ bits, where ℓ_t denotes the length of a binary prefix code of encoding integer t. Specially, when choosing δ code as the prefix code, the proposed schemes can achieve a smaller share size for thresholds $k = 2$ and $k \geq 4$, compared with all currently known constructions. A comparative summary of share sizes in existing evolving threshold secret sharing schemes is provided in Table 1.
- Based on the prefix codes and the properties of polynomial ring, the proposed k-threshold secret sharing schemes have correctness and perfect security. Specially, we present an ingenious mathematical proof demonstrating the

Table 1. Share sizes of evolving k-threshold schemes, where the value of the lowest share size is in bold for each case of k.

threshold	algorithm	share size
$k=2$	[12]	$\ell \cdot O(2^{1+\varepsilon} \lg t)$, $\varepsilon > 0$
	[8]	$\lg t + (\ell+1) \lg\lg t + 4\ell + 1$
	ours	$\boldsymbol{\lfloor \lg t \rfloor + 2\lfloor \lg(\lfloor \lg t \rfloor + 1)\rfloor + \ell}$
$k=3$	[12]	$\ell \cdot O(3^{1+\varepsilon} \lg t)$, $\varepsilon > 0$
	[8]	$2\lg t + 486\ell \lg\lg t \cdot \lg\lg\lg t + 567\ell \lg 3$
	ours	$2\lfloor \lg t \rfloor + 4\lfloor \lg(\lfloor \lg t \rfloor + 1)\rfloor + \ell$
	[10]	$\boldsymbol{\lg t + poly(k) \cdot o(\lg t)}$
$k \geq 4$	[12]	$\ell \cdot O(k^{1+\varepsilon} \lg t)$, $\varepsilon > 0$
	[8]	$(k-1)\lg t + 6k^4\ell \lg\lg t \cdot \lg\lg\lg t + 7k^4\ell \lg k$
	ours	$\boldsymbol{(k-1)\lfloor \lg t \rfloor + 2(k-1)\lfloor \lg(\lfloor \lg t \rfloor + 1)\rfloor + \ell}$

scheme's perfect security. This innovative approach significantly enhances the clarity and comprehensibility of the proof process.

- It is well known that the evolving 2-threshold secret sharing scheme has been studied comprehensively. When $k = 2$, the proposed scheme provides a unified mathematical description for prior evolving 2-threshold secret sharing schemes. In addition, the proposed scheme also achieves the minimal share size for single-bit secret, as stated in [8].

1.2 Technical Overview

We now give a detailed overview of the main techniques we use to obtain our results, explaining the ideas behind each of our constructions. Our approach leverages the algebraic structure of polynomial rings, prefix codes for integers, and innovative proof techniques to achieve a unified and efficient framework for evolving k-threshold secret sharing. Below, we elaborate on the core technical components.

Construction of the Scheme. Our scheme generalizes Shamir's secret sharing to a polynomial ring, enabling a direct and concise construction for evolving k-threshold schemes. The key technical steps are as follows.

Polynomial ring framework. We work over a polynomial ring $\mathbb{F}_p[x]$. For a secret $s \in \mathbb{F}_p[x]$ with degree $\ell - 1$, we construct a polynomial $f(y)$ of the form $f(y) = \sum_{j=0}^{k-2} r_j y^j + s y^{k-1}$ over $\mathbb{F}_p[x]$, where $\{r_j\}_{j=0}^{k-2}$ are random parameters. Then the shares are encoded as evaluations of polynomials. This algebraic structure allows us to naturally extend Shamir's scheme to an unbounded number of participants.

Prefix codes for integers. To handle the evolving nature of the scheme, we map each participant index t to a codeword from a prefix code and ℓ_t denote the length of the corresponding codeword. The corresponding polynomial form y_t derived from its prefix code is then substituted into the pre-constructed polynomial $f(y)$, and the evaluation result of $f(y_t)$ modulo $x^{(k-1)(\ell_t-1)+\ell}$ is assigned as the corresponding share.

Correctness and Secrecy Proofs. The correctness and security of our scheme rely on careful algebraic analysis and combinatorial properties of prefix codes.

Correctness. The scheme ensures correctness through the following mechanism. Upon collecting k shares, we can establish a system of k equations containing k unknown variables $(r_0, r_1, \cdots, r_{k-2}, s)$. By leveraging the algebraic structure of polynomial rings and the properties of prefix codes, we solve the secret s using Gaussian elimination, thereby guaranteeing that any k participants can reconstruct s.

Secrecy. When collecting $k-1$ shares, we can construct a system of $k-1$ equations. To demonstrate security according to the formal definition, we choose two distinct secrets s_0 and s_1, and denote $\{Z_{i_j}^{(s_0)}\}_{j=1}^{k-1}$ and $\{Z_{i_j}^{(s_1)}\}_{j=1}^{k-1}$ the share sets generated under each secret respectively. Given any fixed $k-1$ shares $\{z_{i_j}\}_{j=1}^{k-1}$, the security proof reduces to showing $P(\{Z_{i_j}^{(s_0)} = z_{i_j}\}_{j=1}^{k-1}) = P(\{Z_{i_j}^{(s_1)} = z_{i_j}\}_{j=1}^{k-1})$. Then, we prove the equality $P(\{Z_{i_j}^{(s_0)} = z_{i_j}\}_{j=1}^{k-1}) = P(\{Z_{i_j}^{(s_1)} = z_{i_j}\}_{j=1}^{k-1})$ by demonstrating that the two equation systems possess an equal number of solutions. By leveraging the algebraic structure of polynomials and the properties of prefix codes, along with techniques for solving equations, we rigorously show that these two equation systems indeed contain identical number of solutions. Therefore, the proposed scheme has perfect security.

1.3 Related Works

Shamir [1] and Blakley [2] independently proposed the (k, n) secret sharing scheme in 1979. The construction of Shamir's scheme was based on Lagrange interpolation. Blakley's scheme was established by using the property of points in multidimensional space. In Shamir's scheme, the dealer randomly chose a polynomial in $\mathbb{F}_p[x]$ of degree less than k such that the constant term of this polynomial is the secret and $p > n$. Consequently, the n shares, which are the evaluations of the polynomial at n distinct points, are respectively distributed to n participants. Therefore, each share is an element of $\mathbb{F}_p$ that can be represented with approximately $\lg n$ bits. Later, Karnin et al. [13] proved that the share size of Shamir's scheme is optimal when the length of the secret is $\lg n$ bits. Bogdanov et al. [14] proved that the share size of Shamir's secret sharing scheme is optimal for a 1-bit secret when n is a prime power and $k = n - 1$. Inspired by the Shamir's scheme, many works have constructed secret sharing schemes using different algebraic structures. Cramer and Fehr [15] proposed a black-box secret sharing scheme over any finite Abelian group. Cao et al. [16] proposed a secret sharing scheme based on polynomial rings. Notably, when selecting specific ring structures, the Fast Fourier Transform can be incorporated into the scheme's encoding process, significantly enhancing computational efficiency. This approach established an important framework for subsequent research.

Unlike Shamir [1] uses a finite field to construct a secret sharing scheme, Mignotte [17] and Asmuth et al. [18] independently gave the new constructions of (k, n) threshold secret sharing scheme based on the Chinese Remainder Theorem for integer rings. Both schemes are similar, however, Asmuth-Bloom's scheme improved security compared with Mignotte's scheme. In Asmuth-Bloom's scheme, n integers that are increasing and pairwise coprime are randomly selected, and the product of any k numbers is greater than the value of the secret. Subsequently, the dealer calculated the remainders that module the secret for n integers separately and sent each participant a share consisting of the corresponding integer and its remainder. Due to the superior computational complexity and efficiency of this scheme, it has become a widely studied and highly regarded secret sharing scheme. Then, many works [19–21] utilized the technology of Asmuth-Bloom's scheme to achieve various performances in practical applications.

Due to the significant advantages of secret sharing schemes, they were widely used in other applications, including threshold cryptography [22], multiparty protocol [23] and quantum computation [24]. In 1985, Chor et al. [25] proposed the concept of verifiability for the first time and constructed a verifiable secret sharing scheme. Moreover, Krawczyk's scheme [26] was constructed to achieve computing efficiency with high probability, allowing small statistical errors in security. Ding et al. [27] described a new construction for the communication efficient secret sharing scheme with a small share size to minimize the decoding bandwidth.

However, traditional secret sharing schemes require the dealer to determine the total number of participants n in advance, which cannot adapt to scenarios where new participants dynamically join. To meet practical needs, Komargodski et al. [7, 8] introduced the evolving k-threshold secret sharing scheme, which allows the total number of participants to dynamically increase. Actually, before Komargodski et al. proposed the scheme, Csirmaz and Tardos [28] had conducted preliminary discussions on similar scenarios, but did not provide a concrete construction. In contrast, Komargodski et al.'s contributions [7, 8] not only established a clear definition and design framework for evolving secret sharing but also laid the theoretical foundations for dynamic threshold access structures. On this basis, Komargodski and Paskin-Cherniavsky [29] proposed practical solutions through algebraic operation of detection codes, further promoting the development of evolutionary secret sharing schemes.

The construction of the evolving threshold secret sharing scheme proposed by Komargodski et al. is currently the most commonly used construction method, which establishes a connection between integer prefix codes and evolving 2-threshold secret sharing schemes. This groundbreaking discovery has sparked extensive research interest, and subsequently, numerous scholars have conducted in-depth studies on the evolving 2-threshold schemes based on integer prefix codes. Given a secret with arbitrary length, Okamura et al. [30] first applied the construction of an evolving 2-threshold scheme to more complicated binary codes for positive integers and analyzed the corresponding share size. Furthermore, based on D-ary prefix code for $D \geq 3$, the construction of the evolving 2-threshold scheme was also proposed in [30]. To evaluate the quality of the scheme, Yan et al. [31] introduce a metric to evaluate evolving 2-threshold secret sharing schemes. Based on this, D'Arco et al. [32] proposed a new evolving

2-threshold secret sharing scheme improving on the previous best known scheme and making closer the lower bound.

When $k > 2$, the scheme [8] proposed by Komargodski et al. still exhibits significant limitations. It adopts a hierarchical design by combining a one-time evolving scheme with multiple instances of the Shamir scheme. As the threshold k increases, the number of Shamir scheme invocations grows substantially, leading to a sharp rise in complexity eventually. Later, Alon et al. [33] gave a construction of evolving secret sharing schemes for infinite layered branching programs. Tough their scheme can handle any evolving access structure which can be computed by an layered branching program, but choosing the right generation growth is not trivial. To achieve a simpler construction method for evolving k-threshold secret sharing schemes, Xing and Yuan [12] proposed a novel scheme based on algebraic geometry codes but larger share size than the scheme [8]. To further reduce the share size, D'Arco et al. [10] proposed a new construction of the evolving 3-threshold scheme based on the Chinese Remainder Theorem, reducing the core term of the t-th share size from $\lg t$ to $\lg t$. However, the construction method is a bit complicated and can only be applicable to the case of $k = 3$.

1.4 Organization

The rest of the paper is organized as follows. In Section 2, we review the traditional secret sharing scheme and evolving secret sharing scheme. In Section 3, we first propose a new construction of an evolving 2-threshold secret sharing scheme and elaborate on the security of the scheme. Then, the evolving k-threshold secret sharing scheme is given in Section 4 for $k \geq 3$. In Section 5, we provide a construction of evolving k-threshold secret sharing scheme considering the secret $s \in \{0, 1, \cdots, p-1\}^{\ell}$. Finally, Section 6 concludes the work and discusses the unresolved issues.

2 Models and Notations

Let $\mathbb{N}^+$ denote the set of positive integers. Let $[n] = \{1, 2, \cdots, n\}$ for $n \in \mathbb{N}^+$. Let $|A|$ denote the cardinality of the set A. For any $p \in \mathbb{N}^+$ being a prime, let $\mathbb{F}_p$ denote the finite field with p elements. Let $\mathbb{F}_p[x] = \{\sum_{j=0}^{N} a_j x^j | a_j \in \mathbb{F}_p\}$ as the polynomial ring, where N is a finite positive integer. For any $f(x), g(x) \in \mathbb{F}_p[x]$, if $g(x)$ divides $f(x)$, it can be expressed as $g(x) \mid f(x)$. The result of $g(x)$ dividing $f(x)$ is denoted as *quotient*, which is written as $\frac{f(x)}{g(x)}$. For any $i, a \in \mathbb{N}^+$, $f(x) \in \mathbb{F}_p[x]$, if $x^i \mid f(x)$ with any $i \leq a$ and $x^{a+1} \nmid f(x)$, then a be denoted as *the maximal extractable power* of $f(x)$. For any $d \in \mathbb{N}^+$, let $\mathbb{F}_p[x]/x^d$ be a quotient ring, which is composed of polynomials $f(x) \in \mathbb{F}_p[x]$ with the degree less than d. For any $f(x) \in \mathbb{F}_p[x]/x^d$, if there exists a polynomial $g(x) \in \mathbb{F}_p[x]/x^d$ such that $f(x)g(x) = 1 \in \mathbb{F}_p[x]/x^d$, then $f(x)$ is called to be *invertible*, which be respresented as $f^{-1}(x)$. Equivalently, $f(x)$ has a multiplicative inverse $g(x)$ modulo x^d in the quotient ring $\mathbb{F}_p[x]/x^d$. Let $\mathbb{F}_p[[x]] = \{\sum_{j=0}^{\infty} a_j x^j | a_j \in \mathbb{F}_p\}$.

2.1 Secret Sharing Scheme

Let $\mathcal{P}_n = \{P_1, P_2, \cdots, P_n\}$ represent the set of n participants. The power set of $\mathcal{P}_n$ is written as $2^{\mathcal{P}_n}$. The collection $\mathcal{A} \subseteq 2^{\mathcal{P}_n}$ is monotone if for arbitrary $A \in \mathcal{A}$ and $A \subseteq C$, it holds that $C \in \mathcal{A}$. The access structure is defined as follows.

Definition 1. *An access structure $\mathcal{A} \subseteq 2^{\mathcal{P}_n}$ is a monotone collection of non-empty sets. The set in $\mathcal{A}$ is called qualified, and the set not in $\mathcal{A}$ is called unqualified.*

Definition 2. *For $k, n \in \mathbb{N}^+$ with $k \leq n$, the (k, n)-threshold access structure $\mathcal{A}$ refers to the set containing only all elements in $2^{\mathcal{P}_n}$ of size at least k, i.e.*

$$\mathcal{A} = \{A \in 2^{\mathcal{P}_n} | |A| \geq k\}.$$

Constructing a (k, n) threshold secret sharing scheme requires a dealer, a secret $s \in S$, a set of n participants and a (k, n) access structure $\mathcal{A}$, where S is the domain of the secret. In the scheme, the dealer distributes the shares to all participants such that the shares of any set in $\mathcal{A}$ can correctly recover the secret, while the shares of any set not in $\mathcal{A}$ cannot gain any information about the secret. We denote by $Z_i^{(s)}$ the share of the i-th participant. The formal definition is given as follows.

Definition 3. *A (k, n) threshold secret sharing scheme consists of a pair of algorithms $(\mathcal{E}, \mathcal{R})$, where $\mathcal{E}$ is used to encode the secret s into shares, and $\mathcal{R}$ is used to reconstruct the secret from a set of shares $H \in 2^{\mathcal{P}_n}$. The following requirements shall be satisfied.*

- ***Correctness**. For any qualified set $A \in \mathcal{A}$, the algorithm $\mathcal{R}$ can correctly recover s from the shares of the participants in A, that is*

$$P[\mathcal{R}(\{Z_i^{(s)}\}_{P_i \in A}, A) = s] = 1.$$

- ***Secrecy**. For any unqualified set $C \notin \mathcal{A}$, there is no information about s leaking to the participants in C, that is, for any two different secrets $s_0, s_1 \in S$ and every possible vector of shares $\{z_i\}_{P_i \in C}$, $P(\{Z_i^{(s_0)} = z_i\}_{P_i \in C}) = P(\{Z_i^{(s_1)} = z_i\}_{P_i \in C})$.*

Let $B(Z_i^{(s)})$ represent the bit length of $Z_i^{(s)}$ for $1 \leq i \leq n$. Shamir [1] first proposed a construction for the (k, n)-threshold secret sharing scheme. For an ℓ-bit secret s, the scheme uses the polynomial over $\mathbb{F}_p$ for $p \geq n$. The share size satisfies $B(Z_i^{(s)}) \geq \max\{\ell, \lg p\}$ for any $i \in [n]$ in Shamir's scheme.

2.2 Evolving Secret Sharing Scheme

When the maximum number of n cannot be determined in advance and can even be infinite, conventional secret sharing schemes cannot be applied directly. In this case, evolving secret sharing schemes are developed. We denote $\mathcal{P} = \{P_1, P_2, \cdots, P_n, \cdots\}$ as the set of participants, where $\mathcal{P}$ is possibly infinite. Then, we give some definitions of the evolving secret sharing scheme.

Definition 4. *A collection of sets $\mathcal{A} \subseteq 2^{\mathcal{P}}$ is an evolving access structure if for every $t \in \mathbb{N}^+$, $\mathcal{A}_t :\ = \mathcal{A} \cap 2^{\{P_1, P_2, \cdots, P_t\}}$ is an access structure as defined in Definition 1.*

Definition 5. *For $k \in \mathbb{N}^+$, the evolving k-threshold access structure $\mathcal{A}$ refers to the set consisting only of all elements in $2^{\mathcal{P}}$ of size at least k, i.e.*

$$\mathcal{A} = \{A \in 2^{\mathcal{P}} | |A| \geq k\}.$$

Similarly, constructing an evolving k-threshold secret sharing scheme requires a dealer, a secret $s \in S$, a set of participants and an evolving k-threshold access structure $\mathcal{A}$. The dealer distributes the shares to all participants such that the shares of any set in $\mathcal{A}$ can correctly reconstruct the secret, while the shares of any set not in $\mathcal{A}$ cannot obtain any information about the secret. The formal definition is given as follows.

Definition 6 ([8]). *Let $\mathcal{A}$ be an be an evolving access structure and S be a domain of secrets, where $|S| \geq 2$. A secret sharing scheme for $\mathcal{A}$ and S consists of a pair of algorithms $(\mathcal{E}, \mathcal{R})$, where the sharing procedure $\mathcal{E}$ and the reconstruction procedure $\mathcal{R}$ satisfy the following requirements.*

- ***Share procedure**. $\mathcal{E}(s, \{Z_1^{(s)}, Z_2^{(s)}, \cdots, Z_{j-1}^{(s)}\})$ gets as input a secret $s \in S$ and the secret shares of participants $P_1, P_2, \cdots, P_{j-1}$. It outputs a share for the j-th party. For $j \in \mathbb{N}^+$ and secret shares $Z_1^{(s)}, Z_2^{(s)}, \cdots, Z_{j-1}^{(s)}$ generated for participants $\{P_1, P_2, \cdots, P_{j-1}\}$, respectively, we let*

$$Z_j^{(s)} \leftarrow \mathcal{E}(s, \{Z_1^{(s)}, Z_2^{(s)}, \cdots, Z_{j-1}^{(s)}\}).$$

  be the secret share of participant P_j.
  We abuse notation and sometimes denote by $Z_j^{(s)}$ the random variable that corresponds to the secret share of participant P_j generated as above.
- ***Correctness**. For every $s \in S$ and every $t \in \mathbb{N}^+$, every qualified set in $\mathcal{A}_t$ can reconstruct the secret. That is, for $s \in S$, $t \in \mathbb{N}^+$ and $A \in \mathcal{A}_t$, it holds that*

$$P[\mathcal{R}(\{Z_i^{(s)}\}_{P_i \in A}, A) = s] = 1,$$

  where the probability is over the randomness of the sharing and reconstruction procedures.
- ***Secrecy**. For every $t \in \mathbb{N}^+$, every unqualified set $C \notin \mathcal{A}_t$, and every two secret $s_0, s_1 \in S$, the distribution of the secret shares of participants in C generated with secret s_0 and the distribution of the shares of participants in C generated with secret s_1 are identical. Namely, the distributions $(\{Z_i^{(s_0)} = z_i\}_{P_i \in C})$ and $(\{Z_i^{(s_1)} = z_i\}_{P_i \in C})$ are identical.*

3 Evolving 2-threshold Secret Sharing Scheme over $\mathbb{F}_2[x]$

In this section, we propose an evolving 2-threshold secret sharing scheme for an ℓ-bit secret s. The scheme allows the dealer to distribute the share to every participant such that any single share cannot recover s.

3.1 Proposed Scheme

Before describing the proposed scheme, we first introduce the definition of prefix codes.

Definition 7. *A prefix (or more precisely, prefix-free) code is a code in which no codeword is prefix of any other codeword. A prefix code for the integers is an infinite prefix code $C = \{c_1, c_2, \cdots\}$, where c_i is the codeword of encoding integer i.*

Given a set of binary prefix codes for integers, the codeword of the integer i is denoted as $c_i = (c_{i,0}, c_{i,1}, \cdots, c_{i,\ell_i-1})$, where ℓ_i represents the length of c_i with $i \in \mathbb{N}^+$. The polynomial form of c_i is defined as $y_i = \sum_{j=0}^{\ell_i-1} c_{i,j}x^j \in \mathbb{F}_2[x]$. Given an ℓ-bit binary sequence $s = (s_0, s_1, \cdots, s_{\ell-1})$, the polynomial form of s is defined as $\sum_{j=0}^{\ell-1} s_j x^j \in \mathbb{F}_2[x]$ and the degree is less than ℓ. Then the i-th share is defined as

$$Z_i^{(s)} = r_0 + sy_i \pmod{x^{\ell_i+\ell-1}}, \tag{1}$$

where r_0 is randomly chosen in $\mathbb{F}_2[[x]]$. Notably, s in (1) uses the polynomial form, which would be the default form throughout this paper. As the degree of r_0 is infinite in (1), we cannot choose such $r_0 \in \mathbb{F}_2[[x]]$ in practice. However, due to the operation modulo $x^{\ell_i+\ell-1}$ in (1), the dealer only needs to choose the part of r_0 with a degree less than $\ell_i + \ell - 1$.

For any two participants $P_i, P_j \in \mathcal{P}$ with $i < j$, let $L_{i,j}$ represent the maximal extractable power of $(y_i - y_j)$. The algorithm $\mathcal{R}$ finds out that the following equation

$$s(y_i - y_j) \equiv Z_i^{(s)} - Z_j^{(s)} \pmod{x^{\ell_i+\ell-1}} \tag{2}$$

has a unique solution of s in $\mathbb{F}_2[x]/x^\ell$ as

$$s = \frac{Z_i^{(s)} - Z_j^{(s)}}{x^{L_{i,j}}} \left(\frac{y_i - y_j}{x^{L_{i,j}}}\right)^{-1} \pmod{x^\ell}. \tag{3}$$

The existence and uniqueness of s in $\mathbb{F}_2[x]/x^\ell$ will be provided in the proof of correctness in the next subsection.

Notably, when the dealer generates the i-th share via (1), where i is sufficiently large, the dealer needs to construct the corresponding r_0. Assuming that the dealer has already distributed shares to the first $i-1$ participants. Therefore, the dealer owns the corresponding r_0, which is used to construct the $(i-1)$-th share. The degree of the r_0 is $\ell_{i-1} + \ell - 2$. We need to update r_0 to be a polynomial with degree $\ell_i + \ell - 2$. Then the dealer randomly chooses $\ell_i - \ell_{i-1}$ coefficients from $\mathbb{F}_2$ for $x^{\ell_{i-1}+\ell-1}, \cdots, x^{\ell_i+\ell-2}$ such that the degree of r_0 is extended from $\ell_{i-1} + \ell - 2$ to $\ell_i + \ell - 2$. Hence, the corresponding r_0, which is used to construct the i-th share, is generated.

3.2 Proofs of Correctness and Secrecy

In this subsection, we will prove the correctness and secrecy of the proposed scheme. Before that, we first emphasize several lemmas, which provide useful results for the proofs.

Lemma 1. *Given any finite field $\mathbb{F}_p$, let $f(x), g(x), h(x), k(x) \in \mathbb{F}_p[x]$ with $f(x) \neq 0$ satisfy the congruence equation*

$$f(x)g(x) \equiv f(x)h(x) \pmod{f(x)k(x)}. \tag{4}$$

Then we have

$$g(x) \equiv h(x) \pmod{k(x)}. \tag{5}$$

Proof. According to the definition of congruence equation in (4), there exists $h_1(x) \in \mathbb{F}_p[x]$ such that

$$f(x)g(x) - f(x)h(x) = f(x)k(x)h_1(x), \tag{6}$$

since $\mathbb{F}_p$ is a finite field and $f(x) \neq 0$ in $\mathbb{F}_p[x]$, we can further get

$$g(x) - h(x) = k(x)h_1(x). \tag{7}$$

Therefore, we have

$$g(x) \equiv h(x) \pmod{k(x)}. \tag{8}$$

Lemma 2. *For any $n, k \in \mathbb{N}^+$, let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{F}_p[x]$. If $a_0 \neq 0$, then $f(x)$ is invertible in $\mathbb{F}_p[x]/x^k$.*

Proof. Let $g(x) = x^k$, then we infer that the factor of $g(x)$ must be the form of x^d for any $0 \leq d \leq k$. However, since $f(0) = a_0 \neq 0$, then x^d is not the factor of $f(x)$. Therefore, we have

$$(g(x), f(x)) = 1. \tag{9}$$

According to Euclidean algorithm, there exists $u(x), v(x) \in \mathbb{F}_p[x]$ such that

$$u(x)f(x) + v(x)x^k = 1. \tag{10}$$

Hence, we further derive

$$u(x)f(x) \equiv 1 \pmod{x^k}. \tag{11}$$

Then $u(x)$ modulo x^k is the inverse of $f(x)$ in $\mathbb{F}_p[x]/x^k$.

Lemma 3. *For any $k_1, k_2 \in \mathbb{N}^+$ with $k_1 \leq k_2$, let $f_1(x), g_1(x), f_2(x), g_2(x)$ be polynomials in $\mathbb{F}_p[x]$ satisfying the following congruence equations*

$$\begin{cases} f_1(x) \equiv g_1(x) \pmod{x^{k_1}}, \\ f_2(x) \equiv g_2(x) \pmod{x^{k_2}}. \end{cases} \tag{12}$$

Then we have

$$f_2(x) - f_1(x) \equiv g_2(x) - g_1(x) \pmod{x^{k_1}}. \tag{13}$$

Proof. According to the definition of congruence equations in (12), there exists $h_1(x), h_2(x) \in \mathbb{F}_p[x]$ such that

$$f_1(x) - g_1(x) = x^{k_1} h_1(x), \tag{14}$$

and

$$f_2(x) - g_2(x) = x^{k_2} h_2(x) = x^{k_1} h_2(x) x^{k_2 - k_1}. \tag{15}$$

Subtracting the equation (14) from the equation (15), we can further get

$$(f_2(x) - f_1(x)) - (g_2(x) - g_1(x)) = x^{k_1}(h_2(x)x^{k_2-k_1} - h_1(x)). \tag{16}$$

From (16), we obtain

$$f_2(x) - f_1(x) \equiv g_2(x) - g_1(x) \pmod{x^{k_1}}. \tag{17}$$

Lemma 4. *For any $\ell, \ell_1, k \in \mathbb{N}^+$ with $\ell \leq k$ and $\ell_1 \leq k$, given $f(x), h(x) \in \mathbb{F}_p[x]$ with $f(x) \neq 0$, let ℓ_1 denote the maximal extractable power of $f(x)$. Let $g(x) \in \mathbb{F}_p[x]$ be a unknown polynomial with the degree no more than $\ell - 1$, and $g(x)$ satisfy the following congruence equation*

$$f(x)g(x) \equiv h(x) \pmod{x^k}. \tag{18}$$

Then if $\ell + \ell_1 \leq k$, there exists a unique $g(x) \in \mathbb{F}_p[x]$ with the degree no more than $\ell - 1$ satisfying (18), i.e.

$$g(x) = \frac{h(x)}{x^{\ell_1}} \left(\frac{f(x)}{x^{\ell_1}}\right)^{-1} \pmod{x^\ell}. \tag{19}$$

Proof. As ℓ_1 is the maximal extractable power such that $x^{\ell_1} \mid f(x)$ and $x^{\ell_1+1} \nmid f(x)$, then $f(x)$ can be written $f(x) = x^{\ell_1} f_1(x)$, where $f_1(x) = \frac{f(x)}{x^{\ell_1}} \in \mathbb{F}_p[x]$ and the constant term of $f_1(x)$ is nonzero in $\mathbb{F}_p$. Taking $x^{\ell_1} f_1(x)$ to replace $f(x)$ in (18), then

$$x^{\ell_1} f_1(x) g(x) \equiv h(x) \pmod{x^k}. \tag{20}$$

Since $\ell_1 \leq k$, we infer that x^{ℓ_1} is a factor of $h(x)$, then

$$x^{\ell_1} f_1(x) g(x) \equiv x^{\ell_1} h_1(x) \pmod{x^{\ell_1} x^{k-\ell_1}}, \tag{21}$$

where $h_1(x) = \frac{h(x)}{x^{\ell_1}} \in \mathbb{F}_p[x]$. By using Lemma 1, we can eliminate the common factor x^{ℓ_1} on both sides of (21) and obtain

$$f_1(x) g(x) \equiv h_1(x) \pmod{x^{k-\ell_1}}. \tag{22}$$

As $\ell \leq k - \ell_1$, (22) also holds under modulo x^ℓ, i.e.

$$f_1(x) g(x) \equiv h_1(x) \pmod{x^\ell}. \tag{23}$$

Since the constant term of $f_1(x)$ is nonzero, by using the conclusion of Lemma 2, there exists the inverse of $f_1(x)$ in $\mathbb{F}_p[x]/x^\ell$, then we have

$$g(x) = h_1(x) f_1(x)^{-1} = \frac{h(x)}{x^{\ell_1}} \left(\frac{f(x)}{x^{\ell_1}}\right)^{-1} \pmod{x^\ell}. \tag{24}$$

Therefore, considering in the polynomial ring $\mathbb{F}_p[x]/x^\ell$, there exists a unique solution $g(x)$ satisfying (18).

The Proof of Correctness For $t \in \mathbb{N}^+, A \in \mathcal{A}_t$, then $|A| \geq 2$. We need to show that the ℓ-bit secret s can be correctly reconstructed by the shares of the participants in A.

Without loss of generality, we take two participants P_i and P_j from A with $i < j \leq t$. Since y_i and y_j are the binary prefix codes of i and j, ℓ_i and ℓ_j are the bit length of i and j, thus we have $\ell_i \leq \ell_j$. Then, the i-th and j-th shares are as follows.

$$\begin{cases} Z_i^{(s)} = r_0 + sy_i \pmod{x^{\ell_i+\ell-1}}, \\ Z_j^{(s)} = r_0 + sy_j \pmod{x^{\ell_j+\ell-1}}. \end{cases} \tag{25}$$

By Lemma 3, subtracting the second equation from the first equation in (25), we can further get

$$(y_i - y_j)s \equiv Z_i^{(s)} - Z_j^{(s)} \pmod{x^{\ell_i+\ell-1}}. \tag{26}$$

As c_i is not a prefix of c_j, we have

$$y_i - y_j \neq 0 \pmod{x^{\ell_i}}. \tag{27}$$

Since $L_{i,j}$ denotes the maximal extractable power of $(y_i - y_j)$ satisfying $x^{L_{i,j}} \mid (y_i - y_j)$, by (27), we have

$$L_{i,j} \leq \ell_i - 1. \tag{28}$$

Thus,

$$L_{i,j} + \ell \leq \ell_i + \ell - 1. \tag{29}$$

Using the conclusion of Lemma 4 and combining the bit length of s is ℓ, the congruence equation (26) has a unique solution in $\mathbb{F}_2[x]/x^\ell$, which can be calculated as

$$s = \frac{Z_i^{(s)} - Z_j^{(s)}}{x^{L_{i,j}}} \big(\frac{y_i - y_j}{x^{L_{i,j}}}\big)^{-1} \pmod{x^\ell}. \tag{30}$$

The Proof of Secrecy For any $t \in \mathbb{N}^+, C \notin \mathcal{A}_t$, i.e. $|C| < 2$. We need to prove that the secret s is unable to be recovered by the shares in C. Since the case of $|C| = 0$ is trivial, we only prove the case of $|C| = 1$ in the following.

Assume the only element in C as P_i with $i \leq t$. For any s, we have $Z_i^{(s)} = r_0 + sy_i$ in $\mathbb{F}_2[x]/x^{\ell_i+\ell-1}$. As r_0 is a random variable uniformly distributed in the additive group $\mathbb{F}_2[x]/x^{\ell_i+\ell-1}$, $Z_i^{(s)}$ is independent from sy_i, which makes $Z_i^{(s)}$ independent from s. Hence, $Z_i^{(s)}$ is uniformly random in $\mathbb{F}_2[x]/x^{\ell_i+\ell-1}$ for each selection of s.

Now, we will provide an example to show the processes of distributing shares and reconstructing the secret.

Example. Given a secret $s = 1001$, for the two participants P_3, P_4, let 101 and 11000 be the binary prefix codes of 3 and 4, respectively. According to $\ell + \ell_4 - 1 =$

8, the algorithm $\mathcal{E}$ randomly chooses an 8-bit binary string $r_0 = 10101001$. Then the share of P_3 is given by

$$\begin{aligned} Z_3^{(s)} =& r_0 + sy_3 \pmod{x^6} \\ =& (1+x^2+x^4) + (1+x^3)(1+x^2) \\ =& x^3+x^4+x^5 \pmod{x^6}, \end{aligned}$$

thus, the 3-th share $Z_3^{(s)}$ is 000111.

And the share of P_4 is given by

$$\begin{aligned} Z_4^{(s)} =& r_0 + sy_4 \pmod{x^8} \\ =& (1+x^2+x^4+x^7) + (1+x^3)(1+x) \\ =& x+x^2+x^3+x^7 \pmod{x^8}, \end{aligned}$$

thus, the 4-th share $Z_4^{(s)}$ is 01110001.

Next, we take this example to show how to reconstruct the secret s. Let the bit length of s be 4. Let 101 and 11000 be the prefix codes of 3 and 4, respectively. The shares of the two participants P_3 and P_4 are 000111 and 01110001, respectively. Then we have

$$\begin{cases} Z_3^{(s)} = x^3+x^4+x^5 = r_0 + s(1+x^2) \pmod{x^6}, \\ Z_4^{(s)} = x+x^2+x^3+x^7 = r_0 + s(1+x) \pmod{x^8}. \end{cases}$$

Since $\ell + \ell_3 - 1 = 6$, $y_3 - y_4 = x + x^2 = x(1+x)$ and the bit length of s is 4, the algorithm $\mathcal{R}$ solves the following equation

$$s(y_3 - y_4) \equiv Z_3^{(s)} - Z_4^{(s)} \pmod{x^6}$$

with a unique solution, i.e.

$$\begin{aligned} s =& \frac{Z_3^{(s)} - Z_4^{(s)}}{x} \left(\frac{y_3-y_4}{x}\right)^{-1} \pmod{x^4} \\ =& \frac{x+x^2+x^4+x^5+x^7}{x} \left(\frac{x+x^2}{x}\right)^{-1} \pmod{x^4} \\ =& \frac{1+x+x^3}{1+x} = (1+x+x^3)(1+x)^{-1} \pmod{x^4} \\ \overset{(a)}{=}& (1+x+x^3)(1+x+x^2+x^3) = 1+x^3 \pmod{x^4}. \end{aligned}$$

where (a) holds since $(1+x)(1+x+x^2+x^3) = 1$ in $\mathbb{F}_2[x]/x^4$. Therefore, the algorithm $\mathcal{R}$ outputs $s = 1001$, which is correct.

Share Size. We analyze the share size of the proposed scheme. For the t-th participant, from (1), the share $Z_t^{(s)}$ can be regarded as a polynomial of x in $\mathbb{F}_2$ with the degree no more than $\ell_t + \ell - 2$, where ℓ_t is the length of binary prefix code for the positive integer t. Then, the size of the corresponding share is

$$B(Z_t^{(s)}) = \ell_t + \ell - 1. \tag{31}$$

From (31), we find that it may obtain different share sizes when choosing different binary prefix codes.

Next, we show the corresponding share size by using several binary prefix codes. We denote by $L(\cdot)$ the codeword length for encoding t for $t \in \mathbb{N}^+$. Considering γ code [11], a widely used integer universal code, the codeword length of $\gamma(t)$ is given by

$$L(\gamma(t)) = 2\lfloor \lg t \rfloor + 1.$$

Therefore, when using γ code [11] as the binary prefix code, the size of the t-th share satisfies

$$B(Z_t^{(s)}) = 2\lfloor \lg t \rfloor + \ell. \tag{32}$$

Except for γ code, we also consider another binary prefix code, δ code [11]. The codeword length of $\delta(t)$ is given by

$$L(\delta(t)) = \lfloor \lg t \rfloor + 2\lfloor \lg(\lfloor \lg t \rfloor + 1) \rfloor + 1.$$

Therefore, if using δ code as the binary prefix code, the share size of the t-th satisfies

$$B(Z_t^{(s)}) = \lfloor \lg t \rfloor + 2\lfloor \lg(\lfloor \lg t \rfloor + 1) \rfloor + \ell. \tag{33}$$

In all knowns evolving 2-threshold schemes, the best result of share size is $\lg t + (\ell+1)\lg\lg t + 4\ell + 1$, which is mentioned in [8]. Compared with it, when $\ell = 1$, the result of (33) is approximately equal to the result of [8]. When $\ell \geq 2$, the result is smaller than the result of [8].

4 Evolving k-threshold Secret Sharing Scheme over $\mathbb{F}_2[x]$

Based on similar techniques, we extend the prior scheme to the evolving k-threshold scheme in this section. We will give the construction of the evolving k-threshold scheme over $\mathbb{F}_2[x]$ for an ℓ-bit secret s, where $k \geq 3$. The scheme allows the dealer to distribute the share to each participant such that only no less than k participants can reconstruct s.

4.1 Proposed Scheme

Given a set of binary prefix codes for integers, the codeword of the integer i is denoted as $c_i = (c_{i,0}, c_{i,1}, \cdots, c_{i,\ell_i-1})$, where ℓ_i represents the length of c_i with $i \in \mathbb{N}^+$. The polynomial form of c_i is defined as y_i. Given an ℓ-bit secret s. For any $i \in \mathbb{N}^+$, the i-th share is defined as

$$Z_i^{(s)} = \sum_{j=0}^{k-2} r_j y_i^j + s y_i^{k-1} \pmod{x^{(\ell_i-1)(k-1)+\ell}}, \tag{34}$$

where r_j is randomly chosen from $\mathbb{F}_2[[x]]$ for $0 \leq j \leq k-2$. However, we cannot choose an $r_j \in \mathbb{F}_2[[x]]$ in practice. Due to the operation modulo $x^{(\ell_i-1)(k-1)+\ell}$

in (34), the dealer only needs to choose a part of r_j with degrees less than $(\ell_i-1)(k-1)+\ell$.

Given any k participants $P_{i_1}, P_{i_2}, \cdots, P_{i_k} \in \mathcal{P}$ with $i_1 < \cdots < i_k$, let L_{i_u,i_v} denote the maximal extractable power of $y_{i_u}-y_{i_v}$ for any $u,v \in [k]$ with $u \neq v$. Let $L_{i_m} = (\ell_{i_m}-1)(k-1)+\ell$ for $1 \le m \le k$, and $\alpha = \min_{m=1}^{k}\{L_{i_m} + \sum_{\substack{1\le u<v\le k\\ u,v\neq m}} L_{i_u,i_v}\}$. The algorithm $\mathcal{R}$ finds out that the following equation

$$s\Big(\prod_{1\le u<v\le k}(y_{i_u}-y_{i_v})\Big) \equiv \sum_{m=1}^{k}\prod_{\substack{1\le u<v\le k\\ u,v\neq m}}(y_{i_u}-y_{i_v})Z_{i_m}^{(s)} \pmod{x^{\alpha}} \tag{35}$$

has a unique solution about s in $\mathbb{F}_2[x]/x^{\ell}$ as

$$s = \frac{\sum_{m=1}^{k}\prod_{\substack{1\le u<v\le k\\ u,v\neq m}}(y_{i_u}-y_{i_v})Z_{i_m}^{(s)}}{x^{\sum_{1\le u<v\le k}L_{i_u,i_v}}}\Big(\frac{\prod_{1\le u<v\le k}(y_{i_u}-y_{i_v})}{x^{\sum_{1\le u<v\le k}L_{i_u,i_v}}}\Big)^{-1} \pmod{x^{\ell}}. \tag{36}$$

Similarly, considering in $\mathbb{F}_2[x]/x^{\ell}$, the existence and uniqueness of s will be provided in the proof of correctness in the next subsection.

When the dealer generates the i-th share via (34), where i is sufficiently large, the dealer also needs to construct the corresponding random polynomials $\{r_j\}_{j=0}^{k-2}$. Assume that the dealer has already distributed shares to the first $i-1$ participants. Therefore, the dealer owns the corresponding $\{r_j\}_{j=0}^{k-2}$, which are used to construct the $(i-1)$-th share. For any $0 \le j \le k-2$, the degree of r_j is $(\ell_{i-1}-1)(k-1)+\ell-1$. To obtain the corresponding $\{r_j\}_{j=0}^{k-2}$ used to construct the i-th share, the dealer randomly chooses $(\ell_i-\ell_{i-1})(k-1)$ coefficients in $\mathbb{F}_2$ such that the degree of each r_j is extended from $(\ell_{i-1}-1)(k-1)+\ell-1$ to $(\ell_i-1)(k-1)+\ell-1$. Hence, the corresponding $\{r_j\}_{j=0}^{k-2}$ are constructed.

4.2 Analysis of Share Size

Next, we discuss the corresponding share size. The share size in the proposed evolving k-threshold scheme is analyzed as follows.

For the t-th participant, from (34), the share $Z_t^{(s)}$ is a polynomial of x in $\mathbb{F}_2$ and the degree is $(k-1)(\ell_t-1)+\ell-1$, where ℓ_t is the length of prefix code to encode positive integer t. Hence, the share size satisfies

$$B(Z_t^{(s)}) = (k-1)(\ell_t-1)+\ell. \tag{37}$$

If using γ code as the prefix code, the corresponding share size satisfies

$$B(Z_t^{(s)}) = 2(k-1)\lfloor \lg t\rfloor+\ell. \tag{38}$$

However, when using δ code as the prefix code, then the corresponding share size satisfies

$$B(Z_t^{(s)}) = (k-1)\lfloor \lg t\rfloor + 2(k-1)\lfloor \lg(\lfloor \lg t\rfloor+1)\rfloor+\ell. \tag{39}$$

We tabulate the share sizes of currently known evolving threshold secret sharing schemes. For the proposed scheme, we show the corresponding share size for using δ code as the prefix code. In Table 1, we represent the value of the lowest share size in bold for each case of k. When $k \geq 2$, the proposed schemes can achieve consistently lower share sizes than the scheme given in [8, 12]. When $k = 3$, the proposed scheme's share size is larger than the result of [10]. However, the result of [10] is optimized only for the case $k = 3$.

4.3 Proofs of Correctness and Secrecy

The proofs of the correctness and secrecy of the proposed scheme are given as follows.

The Proof of Correctness For any $t \in \mathbb{N}^+, A \in \mathcal{A}_t$, then $|A| \geq k$, we will prove that the secret s can be correctly recovered by the shares of the participants in A. Since the cases of $|A| \geq k$ include the case of $|A| = k$, we only prove the case of $|A| = k$.

Denote the k elements in A as $P_{i_1}, P_{i_2}, \cdots, P_{i_k}$ with $i_1 < \cdots < i_k \leq t$. Since c_{i_m} is the prefix code of i_m, y_{i_m} is the polynomial form of c_{i_m} and ℓ_{i_m} is the code length of i_m, thus ℓ_{i_m} is increasing on m, where $1 \leq m \leq k$. Recall that the corresponding shares are as follows.

$$\begin{cases} Z_{i_1}^{(s)} = \sum_{j=0}^{k-2} r_j y_{i_1}^j + s y_{i_1}^{k-1} \pmod{x^{(\ell_{i_1}-1)(k-1)+\ell}}, \\ Z_{i_2}^{(s)} = \sum_{j=0}^{k-2} r_j y_{i_2}^j + s y_{i_2}^{k-1} \pmod{x^{(\ell_{i_2}-1)(k-1)+\ell}}, \\ \cdots \\ Z_{i_k}^{(s)} = \sum_{j=0}^{k-2} r_j y_{i_k}^j + s y_{i_k}^{k-1} \pmod{x^{(\ell_{i_k}-1)(k-1)+\ell}}. \end{cases} \tag{40}$$

In order to calculate s, we need to eliminate these elements $r_0, \cdots, r_{k-2}$. Considering each congruence equation in (40), there exists $h_m(x) \in \mathbb{F}_2[x]$ such that the equation

$$\sum_{j=0}^{k-2} r_j y_{i_m}^j + s y_{i_m}^{k-1} - Z_{i_m}^{(s)} = h_m(x) x^{(\ell_{i_m}-1)(k-1)+\ell} \tag{41}$$

holds.
Considering (41), we multiply both sides of the equation by $\prod_{\substack{1 \leq u < v \leq k \\ u,v \neq m}} (y_{i_u} - y_{i_v})$. For the convenience of writing, let $H_{i_m}(x) = \prod_{\substack{1 \leq u < v \leq k \\ u,v \neq m}} (y_{i_u} - y_{i_v})$, and $L_{i_m} = (\ell_{i_m} - 1)(k-1) + \ell$ for $1 \leq m \leq k$, then the equation (41) becomes

$$H_{i_m}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_m}^j + s y_{i_m}^{k-1} - Z_{i_m}^{(s)}\Big) = H_{i_m} h_m(x) x^{L_{i_m}}. \tag{42}$$

Performing the above same steps for each congruent equation in (40), then we can obtain

$$\begin{cases} H_{i_1}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_1}^j + s y_{i_1}^{k-1} - Z_{i_1}^{(s)}\Big) = H_{i_1} h_1(x) x^{L_{i_1}}, \\ H_{i_2}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_2}^j + s y_{i_2}^{k-1} - Z_{i_2}^{(s)}\Big) = H_{i_2} h_2(x) x^{L_{i_2}}, \\ \cdots \\ H_{i_k}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_k}^j + s y_{i_k}^{k-1} - Z_{i_k}^{(s)}\Big) = H_{i_k} h_k(x) x^{L_{i_k}}, \end{cases} \tag{43}$$

Summing all equations in (43), we have

$$\begin{aligned} &\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1} s + \sum_{j=0}^{k-2}\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^j r_j \\ =&\sum_{m=1}^{k} H_{i_m}(x) Z_{i_m}^{(s)} + \sum_{m=1}^{k} H_{i_m}(x) h_m(x) x^{L_{i_m}}. \end{aligned} \tag{44}$$

Consider the coefficient of s as $\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1} = \prod_{\substack{1\le u<v\le k \\ u,v\neq m}} (y_{i_u} - y_{i_v}) y_{i_m}^{k-1}$ in $\mathbb{F}_2[x]$, which is equal to the following Vandermonde determinant, i.e.

$$\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1} \overset{(b)}{=} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ y_{i_1} & y_{i_2} & \cdots & y_{i_k} \\ \vdots & \vdots & \cdots & \vdots \\ y_{i_1}^{k-1} & y_{i_2}^{k-1} & \cdots & y_{i_k}^{k-1} \end{vmatrix} = \prod_{1\le u<v\le k} (y_{i_u} - y_{i_v}), \tag{45}$$

where (b) holds since $\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1}$ is equal to the result of expanding the above determinant based on the last row.

For arbitrary j with $0 \le j \le k-2$, the coefficient of r_j is $\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^j$, then we have

$$\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{j} \overset{(c)}{=} \begin{vmatrix} 1 & 1 & \cdots & 1 \\ y_{i_1} & y_{i_2} & \cdots & y_{i_k} \\ \vdots & \vdots & \cdots & \vdots \\ y_{i_1}^{k-2} & y_{i_2}^{k-2} & \cdots & y_{i_k}^{k-2} \\ y_{i_1}^{j} & y_{i_2}^{j} & \cdots & y_{i_k}^{j} \end{vmatrix} = 0, \tag{46}$$

where (c) holds since $\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^j$ is equal to the result of expanding the above determinant based on the last row.

Taking the results of (45) and (46) into (44), we can further obtain

$$\prod_{1\le u<v\le k} (y_{i_u} - y_{i_v}) s = \sum_{m=1}^{k} H_{i_m}(x) Z_{i_m}^{(s)} + \sum_{m=1}^{k} H_{i_m}(x) h_m(x) x^{L_{i_m}}. \tag{47}$$

Since L_{i_u,i_v} denotes the maximal extractable power of $y_{i_u}-y_{i_v}$ for any $u,v\in[k]$ with $u\neq v$, and $L_{i_m}=(\ell_{i_m}-1)(k-1)+\ell$ for $1\leq m\leq k$, we can infer that each polynomial

$$H_{i_m}(x)h_m(x)x^{L_{i_m}}=\prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}}(y_{i_u}-y_{i_v})x^{L_{i_m}}h_m(x)$$

has the factor x^α for any m, where $\alpha=\min_{1\leq m\leq k}\{L_{i_m}+\sum_{\substack{1\leq u<v\leq k\\ u,v\neq m}}L_{i_u,i_v}\}$.

Substituting $H_{i_m}(x)$ by $\prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}}(y_{i_u}-y_{i_v})$ in (47) and simplying the equation, then we obtain

$$s\Big(\prod_{1\leq u<v\leq k}(y_{i_u}-y_{i_v})\Big)\equiv\sum_{m=1}^{k}\prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}}(y_{i_u}-y_{i_v})Z_{i_m}^{(s)}\pmod{x^\alpha}. \tag{48}$$

We note that the polynomial $\prod_{1\leq u<v\leq k}(y_{i_u}-y_{i_v})$ has the maximal extractable power $\sum_{1\leq u<v\leq k}L_{i_u,i_v}$. If $\sum_{1\leq u<v\leq k}L_{i_u,i_v}+\ell\leq\alpha$, we can use the conclusion of Lemma 4 to recover s. Next, our goal is to prove $\sum_{1\leq u<v\leq k}L_{i_u,i_v}+\ell\leq\alpha$.

Without loss of generality, we assume $\alpha=\min_{1\leq m\leq k}\{L_{i_m}+\sum_{\substack{1\leq u<v\leq k\\ u,v\neq m}}L_{i_u,i_v}\}$ $=L_{i_q}+\sum_{\substack{1\leq u<v\leq k\\ u,v\neq q}}L_{i_u,i_v}$ for some q. Since $L_{i_q}=(\ell_{i_q}-1)(k-1)+\ell$, then we have

$$\begin{aligned}
&\alpha-\sum_{1\leq u<v\leq k}L_{i_u,i_v}-\ell\\
=&L_{i_q}+\sum_{\substack{1\leq u<v\leq k\\ u,v\neq q}}L_{i_u,i_v}-\sum_{1\leq u<v\leq k}L_{i_u,i_v}-\ell\\
=&(\ell_{i_q}-1)(k-1)-\sum_{\substack{1\leq u\leq k\\ u\neq q}}L_{i_u,i_q}\\
=&\sum_{\substack{1\leq u\leq k\\ u\neq q}}(\ell_{i_q}-1-L_{i_u,i_q}).
\end{aligned} \tag{49}$$

For any $u\neq q$ with $1\leq u\leq k$, since $y_{i_u}-y_{i_q}\neq 0$, L_{i_u,i_q} is the maximal extractable power of $y_{i_u}-y_{i_q}$, and ℓ_{i_q} is the prefix codeword length of y_{i_q}, then we have $L_{i_u,i_q}\leq\ell_{i_q}-1$. Replacing the result in (49), we get

$$\sum_{1\leq u<v\leq k}L_{i_u,i_v}+\ell\leq\alpha. \tag{50}$$

By using the conclusion of Lemma 4, we can recover the unique solution about s of (48) in $\mathbb{F}_2[x]/x^\ell$ as

$$s=\frac{\sum_{m=1}^{k}\prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}}(y_{i_u}-y_{i_v})Z_{i_m}^{(s)}}{x^{\sum_{1\leq u<v\leq k}L_{i_u,i_v}}}\Big(\frac{\prod_{1\leq u<v\leq k}(y_{i_u}-y_{i_v})}{x^{\sum_{1\leq u<v\leq k}L_{i_u,i_v}}}\Big)^{-1}\pmod{x^\ell}. \tag{51}$$

Therefore, the correctness of the proposed scheme has been proved completely.

In the second part of this subsection, we demonstrate the security of this scheme. Before proving the security, we first present a lemma that provides a useful conclusion for proving the security.

Lemma 5. *Let $i_1, i_2, \cdots, i_{k-1} \in \mathbb{N}^+$ satisfy $i_1 < i_2 < \cdots < i_{k-1}$, c_{i_m} represent the p-ary codeword of encoding the integer i_m with a p-ary prefix code and ℓ_{i_m} denote the codeword length of c_{i_m}, where $1 \leq m \leq k-1$. Let y_{i_m} represent the corresponding polynomial form of c_{i_m}. For $\ell \in \mathbb{N}^+$, let $s_0, s_1 \in \mathbb{F}_p[x]/x^{\ell}$ be two different polynomials. Given $k-1$ polynomials $z_{i_m} \in \mathbb{F}_p[x]/x^{L_{i_m}}$ for $1 \leq m \leq k-1$, where $L_{i_m} = (\ell_{i_m} - 1)(k-1) + \ell$, let $(r_{0,0}, r_{0,1}, \cdots, r_{0,k-2})$ and $(r_{1,0}, r_{1,1}, \cdots, r_{1,k-2})$ respectively be the unknowns of the following two congruence equations*

$$\begin{cases} z_{i_1} = \sum_{j=0}^{k-2} r_{0,j} y_{i_1}^j + s_0 y_{i_1}^{k-1} \pmod{x^{L_{i_1}}}, \\ z_{i_2} = \sum_{j=0}^{k-2} r_{0,j} y_{i_2}^j + s_0 y_{i_2}^{k-1} \pmod{x^{L_{i_2}}}, \\ \qquad\cdots \\ z_{i_{k-1}} = \sum_{j=0}^{k-2} r_{0,j} y_{i_{k-1}}^j + s_0 y_{i_{k-1}}^{k-1} \pmod{x^{L_{i_{k-1}}}}, \end{cases} \tag{52}$$

and

$$\begin{cases} z_{i_1} = \sum_{j=0}^{k-2} r_{1,j} y_{i_1}^j + s_1 y_{i_1}^{k-1} \pmod{x^{L_{i_1}}}, \\ z_{i_2} = \sum_{j=0}^{k-2} r_{1,j} y_{i_2}^j + s_1 y_{i_2}^{k-1} \pmod{x^{L_{i_2}}}, \\ \qquad\cdots \\ z_{i_{k-1}} = \sum_{j=0}^{k-2} r_{1,j} y_{i_{k-1}}^j + s_1 y_{i_{k-1}}^{k-1} \pmod{x^{L_{i_{k-1}}}}, \end{cases} \tag{53}$$

where $r_{h,j} \in \mathbb{F}_p[x]/x^{L_{i_{k-1}}}$ for $0 \leq h \leq 1$, $0 \leq j \leq k-2$.

Then the equations (52) and (53) have the same number of solutions.

Proof. We prove the conclusion by contradiction. Without loss of generality, we assume that the congruence equation (52) has N_0 solutions and the congruence equation (53) has N_1 solutions with $N_0 > N_1$. Let $\mathbb{R}_0$ and $\mathbb{R}_1$ respectively be the solution sets of (52) and (53). According to the definition of congruence equations in (52), there exists $w_{0,j}$ satisfying following equation

$$\begin{cases} z_{i_1} = \sum_{j=0}^{k-2} r_{0,j} y_{i_1}^j + s_0 y_{i_1}^{k-1} + w_{0,1} x^{L_{i_1}}, \\ z_{i_2} = \sum_{j=0}^{k-2} r_{0,j} y_{i_2}^j + s_0 y_{i_2}^{k-1} + w_{0,2} x^{L_{i_2}}, \\ \qquad\cdots \\ z_{i_{k-1}} = \sum_{j=0}^{k-2} r_{0,j} y_{i_{k-1}}^j + s_0 y_{i_{k-1}}^{k-1} + w_{0,k-1} x^{L_{i_{k-1}}}, \end{cases} \tag{54}$$

where $w_{0,m} \in \mathbb{F}_p[x]/x^{(\ell_{i_{k-1}}-1)(2k-3)+\ell-L_{i_m}}$ for $1 \leq m \leq k-1$.

Correspondingly, there exists $w_{1,m} \in \mathbb{F}_p[x]/x^{(\ell_{i_{k-1}}-1)(2k-3)+\ell-L_{i_m}}$ such that congruence equation (53) satisfys the following form

$$\begin{cases} z_{i_1} = \sum_{j=0}^{k-2} r_{1,j} y_{i_1}^j + s_1 y_{i_1}^{k-1} + w_{1,1} x^{L_{i_1}}, \\ z_{i_2} = \sum_{j=0}^{k-2} r_{1,j} y_{i_2}^j + s_1 y_{i_2}^{k-1} + w_{1,2} x^{L_{i_2}}, \\ \qquad\cdots \\ z_{i_{k-1}} = \sum_{j=0}^{k-2} r_{1,j} y_{i_{k-1}}^j + s_1 y_{i_{k-1}}^{k-1} + w_{1,k-1} x^{L_{i_{k-1}}}. \end{cases} \tag{55}$$

It finds that the number of solutions $(r_{0,0}, r_{0,1}, \cdots, r_{0,k-2}, w_{0,1}, w_{0,2}, \cdots, w_{0,k-1})$ of (54) and the number of solutions $(r_{0,0}, r_{0,1}, \cdots, r_{0,k-2})$ of (52) are same, the numbers of solutions of (55) and (53) are same. Thus, we can prove that (52)) and (53) have the same number of solutions, to show that the conclusion holds. Next, we analyze the solution cases of (54) and (55). According to the assumptions, (52) has solutions. As $\mathbb{R}_0$ is the solution set of (52), we choose one of solutions in $\mathbb{R}_0$ and denote it as $R_0^1 = (r_{0,0}^1, r_{0,1}^1, \cdots, r_{0,k-2}^1)$, then $R_{0,+}^1 = (r_{0,0}^1, r_{0,1}^1, \cdots, r_{0,k-2}^1, w_{0,1}, w_{0,2}, \cdots, w_{0,k-1})$ is the solution of (54), where $w_{0,m} = \frac{\sum_{j=0}^{k-2} r_{0,j}^1 y_{i_m}^j + s_0 y_{i_m}^{k-1} - z_{i_m}}{x^{L_{i_m}}}$ for $1 \le m \le k-1$.

Consider the quotient ring $\mathbb{F}_p[x]/x^{(\ell_{i_{k-1}}-1)(2k-3)+\ell}$. Subtracting (55) from (54), then we have

$$\begin{cases} (s_0 - s_1) y_{i_1}^{k-1} = \sum_{j=0}^{k-2} r_j y_{i_1}^j + w_1 x^{L_{i_1}}, \\ (s_0 - s_1) y_{i_2}^{k-1} = \sum_{j=0}^{k-2} r_j y_{i_2}^j + w_2 x^{L_{i_2}}, \\ \qquad\qquad \cdots \\ (s_0 - s_1) y_{i_{k-1}}^{k-1} = \sum_{j=0}^{k-2} r_j y_{i_{k-1}}^j + w_{k-1} x^{L_{i_{k-1}}}, \end{cases} \tag{56}$$

where $r_j = r_{1,j} - r_{0,j}$ for $0 \le j \le k-2$ and $w_m = w_{1,m} - w_{0,m}$ for $1 \le m \le k-1$.

Suppose that there exists a solution $R = (r_0, \cdots, r_{k-2}, w_1, \cdots, w_{k-1})$ of (56). Then combining R with the solution $R_{0,+}^1$ of (54), we assert that $R_{0,+}^1 + R$ is the solution of (55). Let $R' = (r_0, \cdots, r_{k-2})$ be the part of R, combining that $R_0^1 = (r_{0,0}^1, r_{0,1}^1, \cdots, r_{0,k-2}^1)$ is the solution of (52), then $R_0^1 + R' = (r_{0,0}^1 + r_0, r_{0,1}^1 + r_1, \cdots, r_{0,k-2}^1 + r_{k-2})$ is a solution of (53). When R_0^1 is taken across the whole solution set $\mathbb{R}_0$, (53) has at least N_0 different solutions, which contradicts that (53) has N_1 solutions. Therefore, (52) and (53) have the same number of solutions. Next, our goal is to show that (56) has solutions.

We first consider the following equations

$$\begin{cases} (s_0 - s_1) y_{i_1}^{k-1} = \sum_{j=0}^{k-2} r_j y_{i_1}^j, \\ (s_0 - s_1) y_{i_2}^{k-1} = \sum_{j=0}^{k-2} r_j y_{i_2}^j, \\ \qquad\qquad \cdots \\ (s_0 - s_1) y_{i_{k-1}}^{k-1} = \sum_{j=0}^{k-2} r_j y_{i_{k-1}}^j. \end{cases} \tag{57}$$

Let $r = (r_0, r_1, \cdots, r_{k-2})$ be unknowns. Call that a fact, for a monic polynomial $f(y) = y^{k-1} + \sum_{i=1}^{k-1} a_i y^{k-1-i}$ with known $k-1$ roots $(y_1, y_2, \cdots, y_{k-1})$, then the $k-1$ coefficients $(a_1, a_2, \cdots, a_{k-1})$ are uniquely determined. By the formulas of the relation between roots and coefficients, we can calculate $a_i = (-1)^i \sigma_i(y_1, y_2, \cdots, y_{k-1}) = \sum_{\substack{C \subseteq [k-1] \\ |C|=i}} \prod_{g \in C} y_g$. Thus, we can obtain a solution about r in (57), i.e.

$$\begin{aligned} r_j =& (-1)^{k-2-j} (s_0 - s_1) \sigma_{k-1-j}(y_{i_1}, y_{i_2}, \cdots, y_{i_{k-1}}) \\ =& (-1)^{k-2-j} (s_0 - s_1) \sum_{\substack{C \subseteq [k-1] \\ |C| = k-1-j}} \prod_{g \in C} y_{i_g} \end{aligned} \tag{58}$$

for $0 \leq j \leq k-2$.

From above, we can construct a solution of (56) as

$$R = (\{r_j = (-1)^{k-2-j}(s_0 - s_1) \sum_{\substack{C \subseteq [k-1] \\ |C|=k-1-j}} \prod_{g \in C} y_{i_g}\}_{j=0}^{k-2}, \{w_m = 0\}_{m=1}^{k-1}). \quad (59)$$

Thus, the proof is completed.

The Proof of Secrecy For any $t \in \mathbb{N}^+, C \notin \mathcal{A}_t$, we prove that the secret s is unable to be recovered by the shares of participants in C. C is unqualified, then $|C| < k$. Since the cases of $|C| < k$ include the case of $|C| = k-1$, we only prove the case of $|C| = k-1$. Other cases can be proved accordingly.

Denote the elements in C as $P_{i_1}, \cdots, P_{i_{k-1}}$ with $i_1 < \cdots < i_{k-1} \leq t$. We choose any two distinct secrets s_0, s_1, let $Z_{i_m}^{(s_0)}$ and $Z_{i_m}^{(s_1)}$ be corresponding the i_m-th shares for $1 \leq m \leq k-1$. We need to prove that for any $z_{i_m} \in \mathbb{F}_2[x]/x^{L_{i_m}}$, where $L_{i_m} = (\ell_{i_m} - 1)(k-1) + \ell$ for $1 \leq m \leq k-1$, the following two probabilities are equal, i.e.

$$P(\{Z_{i_m}^{(s_0)} = z_{i_m}\}_{m=1}^{k-1}) = P(\{Z_{i_m}^{(s_1)} = z_{i_m}\}_{m=1}^{k-1}). \quad (60)$$

We first analyze the value of $P(\{Z_{i_m}^{(s_0)} = z_{i_m}\}_{m=1}^{k-1})$. Consider the following congruence equations

$$\begin{cases} z_{i_1} = \sum_{j=0}^{k-2} r_{0,j} y_{i_1}^j + s_0 y_{i_1}^{k-1} \pmod{x^{L_{i_1}}}, \\ z_{i_2} = \sum_{j=0}^{k-2} r_{0,j} y_{i_2}^j + s_0 y_{i_2}^{k-1} \pmod{x^{L_{i_2}}}, \\ \qquad\cdots \\ z_{i_{k-1}} = \sum_{j=0}^{k-2} r_{0,j} y_{i_{k-1}}^j + s_0 y_{i_{k-1}}^{k-1} \pmod{x^{L_{i_{k-1}}}}. \end{cases} \quad (61)$$

Though $r_{0,j} \in \mathbb{F}_2[[x]]$, only the part with degree less than $L_{i_{k-1}}$ participates in the above calculation. Hence, we only need to consider the part of $r_{0,j}$ modulo $x^{L_{i_{k-1}}}$. Therefore, the whole space of the solution vector $(r_{0,0}, r_{0,1}, \cdots, r_{0,k-2})$ can be regarded as $\{\mathbb{F}_2[x]/x^{L_{i_{k-1}}}\}^{k-1}$. The value of $P(\{Z_{i_m}^{(s_0)} = z_{i_m}\}_{m=1}^{k-1})$ is equal to the ratio of the number of solution $(r_{0,0}, r_{0,1}, \cdots, r_{0,k-2})$ of (61) in the whole space $\{\mathbb{F}_2[x]/x^{L_{i_{k-1}}}\}^{k-1}$.

By similar analysis, the value of $P(\{Z_{i_m}^{(s_1)} = z_{i_m}\}_{m=1}^{k-1})$ is equal to the ratio of the number of solution of (62) in the whole space $\{\mathbb{F}_2[x]/x^{L_{i_{k-1}}}\}^{k-1}$.

$$\begin{cases} z_{i_1} = \sum_{j=0}^{k-2} r_{1,j} y_{i_1}^j + s_1 y_{i_1}^{k-1} \pmod{x^{L_{i_1}}}, \\ z_{i_2} = \sum_{j=0}^{k-2} r_{1,j} y_{i_2}^j + s_1 y_{i_2}^{k-1} \pmod{x^{L_{i_2}}}, \\ \qquad\cdots \\ z_{i_{k-1}} = \sum_{j=0}^{k-2} r_{1,j} y_{i_{k-1}}^j + s_1 y_{i_{k-1}}^{k-1} \pmod{x^{L_{i_{k-1}}}}. \end{cases} \quad (62)$$

By Lemma 5, the numbers of solutions of the congruence equations (61) and (62) are same, which implies $P(\{Z_{i_m}^{(s_0)} = z_{i_m}\}_{m=1}^{k-1}) = P(\{Z_{i_m}^{(s_1)} = z_{i_m}\}_{m=1}^{k-1})$. Then, the security of the proposed scheme is completely proved.

5 Construction of Evolving k-threshold Secret Sharing Scheme over A Polynomial Ring $\mathbb{F}_p[x]$

As described in the prior sections, for an ℓ-bit binary sequence s, based on binary prefix codes, we have proposed the constructions of evolving k-threshold secret sharing scheme over $\mathbb{F}_2[x]$, where $k \geq 2$. When consider the secret $s \in \{0, 1, \cdots, p-1\}^\ell$ for any prime $p \in \mathbb{N}^+$, based on p-ary prefix codes, we extend the proposed evolving k-threshold secret sharing scheme to $\mathbb{F}_p[x]$.

5.1 Proposed Scheme

Given a set of p-ary prefix codes for positive integers, the codeword of the integer i is denoted as $c_i = (c_{i,0}, c_{i,1}, \cdots, c_{i,\ell_i-1})$, where $c_{i,j} \in \mathbb{F}_p$ for $0 \leq j \leq \ell_i - 1$ and ℓ_i denotes the codeword length of c_i. The polynomial form of c_i is defined as $y_i = \sum_{j=0}^{j=\ell_i-1} c_{i,j} x^j \in \mathbb{F}_p[x]$. Given an ℓ length p-ary sequence $s = (s_0, s_1, \cdots, s_{\ell-1})$, the polynomial form of s is defined as $\sum_{j=0}^{\ell-1} s_j x^j \in \mathbb{F}_p[x]$ and the degree is less than ℓ. For any $i \in \mathbb{N}^+$, then the i-th share is defined as

$$Z_i^{(s)} = \sum_{j=0}^{k-2} r_j {y_i}^j + s {y_i}^{k-1} \pmod{x^{(\ell_i-1)(k-1)+\ell}}, \tag{63}$$

where r_j is randomly chosen from $\mathbb{F}_p[[x]]$ for $0 \leq j \leq k-2$.

Given any k participants $P_{i_1}, P_{i_2}, \cdots, P_{i_k} \in \mathcal{P}$ with $i_1 < \cdots < i_k$, let L_{i_u,i_v} denote the maximal extractable power of $y_{i_u} - y_{i_v}$ satisfying $x^{L_{i_u,i_v}} \mid (y_{i_u} - y_{i_v})$, for any $u, v \in [k]$ with $u \neq v$. Let $L_{i_m} = (\ell_{i_m} - 1)(k-1) + \ell$ for $1 \leq m \leq k$ and $\alpha = \min_{1\leq m\leq k}\{L_{i_m} + \sum_{\substack{1\leq u<v\leq k\\ u,v\neq m}} L_{i_u,i_v}\}$. The algorithm $\mathcal{R}$ finds out that the following equation

$$s\Big(\prod_{1\leq u<v\leq k} (y_{i_u} - y_{i_v})\Big) \equiv \sum_{m=1}^{k} (-1)^{m-1} \prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}} (y_{i_u} - y_{i_v}) Z_{i_m}^{(s)} \pmod{x^\alpha} \tag{64}$$

has a unique solution about s in $\mathbb{F}_p[x]/x^\ell$ as

$$\begin{aligned} s =& \frac{\sum_{m=1}^{k} (-1)^{m-1} \prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}} (y_{i_u} - y_{i_v}) Z_{i_m}^{(s)}}{x^{\sum_{1\leq u<v\leq k} L_{i_u,i_v}}} \\ &\times \Big(\frac{\prod_{1\leq u<v\leq k}(y_{i_u} - y_{i_v})}{x^{\sum_{1\leq u<v\leq k} L_{i_u,i_v}}}\Big)^{-1} \pmod{x^\ell}. \end{aligned} \tag{65}$$

The existence and uniqueness of s will be provided in the proof of correctness in next subsection.

Now, we will provide an example to show the processes of distributing shares and reconstructing the secret.

Example. In this example, we choose $k=3$, $p=3$ the secret $s=2101$ with $\ell=4$. For the three participants P_2, P_5 and P_8, let 01, 102 and 112 be the prefix codes of 2, 5 and 8, respectively. As $\ell+(k-1)(\ell_8-1)=8$, the algorithm $\mathcal{E}$ randomly chooses two 8-bit binary strings $r_0=01201200$ and $r_1=20100010$, then the share of P_2 is given by

$$\begin{aligned}Z_2^{(s)}=&r_0+r_1y_2+sy_2{}^2\\=&(x+2x^2+x^4+2x^5)+(2+x^2)x+(2+x+x^3)x^2\\=&x^2+2x^3+x^4 \pmod{x^6},\end{aligned}$$

thus, the share $Z_2^{(s)}$ is 001210.

The share of P_5 is calculated by

$$\begin{aligned}Z_5^{(s)}=&r_0+r_1y_5+sy_5{}^2\\=&(x+2x^2+x^4+2x^5)+(2+x^2+x^6)(1+2x^2)+(2+x+x^3)(1+2x^2)^2\\=&1+2x+2x^3+2x^4+x^5+x^6+x^7 \pmod{x^8},\end{aligned}$$

thus, the share $Z_5^{(s)}$ is 12022111.

And the share of P_8 is given by

$$\begin{aligned}Z_8^{(s)}=&r_0+r_1y_8+sy_8{}^2\\=&(x+2x^2+x^4+2x^5)+(2+x^2+x^6)(1+x+2x^2)\\&+(2+x+x^3)(1+x+2x^2)^2\\=&1+2x+x^2+2x^4+2x^5+2x^6+2x^7 \pmod{x^8},\end{aligned}$$

thus, the share $Z_8^{(s)}$ is 12102222.

Let 01, 102 and 112 be the prefix codes of 2, 5 and 8, respectively. Let 001210, 12022111 and 12102222 respectively be the shares of the three participants P_2, P_5, P_8. Then we have

$$\begin{cases}Z_{P_2}^{(s)}=x^2+2x^3+x^4 \pmod{x^6},\\Z_{P_5}^{(s)}=1+2x+2x^3+2x^4+x^5+x^6+x^7 \pmod{x^8},\\Z_{P_8}^{(s)}=1+2x+x^2+2x^4+2x^5+2x^6+2x^7 \pmod{x^8}.\end{cases}$$

As $(y_2-y_5)(y_2-y_8)(y_5-y_8)=x(2+x+2x^2+2x^3+2x^4)$ and the bit length of the secret s is 4, the algorithm $\mathcal{R}$ recovers the secret s as

$$\begin{aligned}s=&\frac{\sum_{m=1}^{3}(-1)^{m-1}\prod_{\substack{1\le u<v\le 3\\u,v\neq m}}(y_{i_u}-y_{i_v})Z_{i_m}^{(s)}}{x^{\sum_{1\le u<v\le 3}L_{i_u,i_v}}}\left(\frac{\prod_{1\le u<v\le 3}(y_{i_u}-y_{i_v})}{x^{\sum_{1\le u<v\le 3}L_{i_u,i_v}}}\right)^{-1} \pmod{x^4}\\=&\frac{Z_2^{(s)}(y_5-y_8)-Z_5^{(s)}(y_2-y_8)+Z_8^{(s)}(y_2-y_5)}{x^{\sum_{1\le u<v\le 3}L_{i_u,i_v}}}\left(\frac{(y_2-y_5)(y_2-y_8)(y_5-y_8)}{x^{\sum_{1\le u<v\le 3}L_{i_u,i_v}}}\right)^{-1}\\=&\frac{x(1+x+2x^2+2x^3+x^4+x^5+2x^6+x^8)}{x}\left(\frac{x(2+x+2x^2+2x^3+2x^4)}{x}\right)^{-1}\end{aligned}$$

$$
\begin{aligned}
=&(1+x+2x^2+2x^3)(2+x+2x^2+2x^3)^{-1} \pmod{x^4}\\
=&(1+x+2x^2+2x^3)(2+2x+2x^3) \pmod{x^4}\\
=&2+x+x^3 \pmod{x^4},
\end{aligned}
$$

thus, the secret is 2101.

5.2 Proofs of Correctness and Secrecy

We will prove the correctness and secrecy of the proposed scheme in this subsection.

The proof of Correctness For any $t \in \mathbb{N}^+, A \in \mathcal{A}_t$, we will prove that the secret s can be correctly recovered by the shares of the participants in A. Similarly, we only need to consider the case of $|A| = k$ when $|A| \geq k$.

Using the similar proof method proposed in Subsection 4.3, we make slight modifications to the proof. Denote the k elements in A as $P_{i_1}, P_{i_2}, \cdots, P_{i_k}$ with $i_1 < \cdots < i_k \leq t$, the corresponding shares are as follows.

$$
\begin{cases}
Z_{i_1}^{(s)} = \sum_{j=0}^{k-2} r_j y_{i_1}^j + s y_{i_1}^{k-1} \pmod{x^{(\ell_{i_1}-1)(k-1)+\ell}},\\
Z_{i_2}^{(s)} = \sum_{j=0}^{k-2} r_j y_{i_2}^j + s y_{i_2}^{k-1} \pmod{x^{(\ell_{i_2}-1)(k-1)+\ell}},\\
\qquad\qquad\cdots\\
Z_{i_k}^{(s)} = \sum_{j=0}^{k-2} r_j y_{i_k}^j + s y_{i_k}^{k-1} \pmod{x^{(\ell_{i_k}-1)(k-1)+\ell}}.
\end{cases}
\tag{66}
$$

Considering the congruence equations in (66), there exists $h_m(x) \in \mathbb{F}_p[x]$ satisfying the equation

$$
\sum_{j=0}^{k-2} r_j y_{i_m}^j + s y_{i_m}^{k-1} - Z_{i_m}^{(s)} = h_m(x) x^{(\ell_{i_m}-1)(k-1)+\ell}. \tag{67}
$$

Then, we multiply both sides of the equation (67) by $H_{i_m}(x)$, where $H_{i_m}(x) = (-1)^{m-1}\prod_{\substack{1\leq u<v\leq k\\ u,v\neq m}}(y_{i_u} - y_{i_v})$. Let $L_{i_m} = (\ell_{i_m}-1)(k-1)+\ell$ for $1 \leq m \leq k$. Performing the above steps for each congruent equation in (66), then, we have

$$
\begin{cases}
H_{i_1}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_1}^j + s y_{i_1}^{k-1} - Z_{i_1}^{(s)}\Big) = H_{i_1} h_1(x) x^{L_{i_1}},\\
H_{i_2}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_2}^j + s y_{i_2}^{k-1} - Z_{i_2}^{(s)}\Big) = H_{i_2} h_2(x) x^{L_{i_2}},\\
\qquad\qquad\cdots\\
H_{i_k}(x)\Big(\sum_{j=0}^{k-2} r_j y_{i_k}^j + s y_{i_k}^{k-1} - Z_{i_k}^{(s)}\Big) = H_{i_k} h_k(x)(x) x^{L_{i_k}},
\end{cases}
\tag{68}
$$

Summing these equations, we can obtain

$$
\begin{aligned}
&\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1} s + \sum_{j=0}^{k-2}\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^j r_j\\
=&\sum_{m=1}^{k} H_{i_m}(x) Z_{i_m}^{(s)} + \sum_{m=1}^{k} H_{i_m}(x) h_m(x) x^{(\ell_{i_m}-1)(k-1)+\ell}.
\end{aligned}
\tag{69}
$$

Consider the coefficient of s as $\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1}$ in $\mathbb{F}_p[x]$, which is equal to the following Vandermonde determinant, i.e.

$$\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{k-1} = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ y_{i_1} & y_{i_2} & \cdots & y_{i_k} \\ \vdots & \vdots & \cdots & \vdots \\ y_{i_1}^{k-1} & y_{i_2}^{k-1} & \cdots & y_{i_k}^{k-1} \end{vmatrix} = \prod_{1\le u<v\le k} (y_{i_u} - y_{i_v}). \tag{70}$$

For arbitrary j with $0 \le j \le k-2$, the coefficient of r_j is $\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{j}$ in $\mathbb{F}_p[x]$, then we have

$$\sum_{m=1}^{k} H_{i_m}(x) y_{i_m}^{j} = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ y_{i_1} & y_{i_2} & \cdots & y_{i_k} \\ \vdots & \vdots & \cdots & \vdots \\ y_{i_1}^{k-2} & y_{i_2}^{k-2} & \cdots & y_{i_k}^{k-2} \\ y_{i_1}^{j} & y_{i_2}^{j} & \cdots & y_{i_k}^{j} \end{vmatrix} = 0 \tag{71}$$

for $0 \le j \le k-2$. Taking the results of (70) and (71) into (69), we can further obtain

$$\prod_{1\le u<v\le k} (y_{i_u} - y_{i_v}) s = \sum_{m=1}^{k} H_{i_m}(x) Z_{i_m}^{(s)} + \sum_{m=1}^{k} H_{i_m}(x) h_m(x) x^{(\ell_{i_m}-1)(k-1)+\ell}. \tag{72}$$

As $\alpha = \min_{1\le m\le k}\{L_{i_m} + \sum_{\substack{1\le u<v\le k \\ u,v\ne m}} L_{i_u,i_v}\}$ and $L_{i_m} = (\ell_{i_m}-1)(k-1)+\ell$ for $1 \le m \le k$, then that each polynomial

$$H_{i_m}(x) h_m(x) x^{L_{i_m}} = \prod_{\substack{1\le u<v\le k \\ u,v\ne m}} (y_{i_u} - y_{i_v}) x^{L_{i_m}} h_m(x)$$

has the factor x^{α} for any m. Therefore, (72) can be derived into

$$\prod_{1\le u<v\le k} (y_{i_u} - y_{i_v}) s \equiv \sum_{m=1}^{k} (-1)^{m-1} \prod_{\substack{1\le u<v\le k \\ u,v\ne m}} (y_{i_u} - y_{i_v}) Z_{i_m}^{(s)} \pmod{x^{\alpha}}. \tag{73}$$

As $\prod_{1\le u<v\le k}(y_{i_u} - y_{i_v})$ has the maximal extractable power $\sum_{1\le u<v\le k} L_{i_u,i_v}$, and the codeword length of s is ℓ, we get $\sum_{1\le u<v\le k} L_{i_u,i_v} + \ell \le \alpha$ (the proof is proposed in Subsection 4.3). Using Lemma 4, we can recover the unique solution of s in $\mathbb{F}_p[x]/x^{\ell}$, which can be written as

$$\begin{aligned} s = & \frac{\sum_{m=1}^{k} (-1)^{m-1} \prod_{\substack{1\le u<v\le k \\ u,v\ne m}} (y_{i_u} - y_{i_v}) Z_{i_m}^{(s)}}{x^{\sum_{1\le u<v\le k} L_{i_u,i_v}}} \\ & \times \Big(\frac{\prod_{1\le u<v\le k} (y_{i_u} - y_{i_v})}{x^{\sum_{1\le u<v\le k} L_{i_u,i_v}}} \Big)^{-1} \pmod{x^{\ell}}. \end{aligned} \tag{74}$$

As for the security of the proposed scheme, a similar proof method has been mentioned in Subsection 4.3. Therefore, we will no longer describe it here.

Table 2. Comparision of evolving k-threshold schemes based on p-ary prefix codes.

threshold	scheme	$D(Z_t^{(s)})$
$k=2$	[30]	$(\lfloor \log_p t \rfloor + 2\lfloor \log_p(\lfloor \log_p t \rfloor + 1)\rfloor + 2) \cdot \max\{\lceil \lg(p+1) \rceil, \ell\}$
	ours	$\boldsymbol{\lfloor \log_p t \rfloor + 2\lfloor \log_p(\lfloor \log_p t \rfloor + 1)\rfloor + 1 + \ell}$
$k \geq 3$	none	/
	ours	$\boldsymbol{(k-1)(\lfloor \log_p t \rfloor + 2\lfloor \log_p(\lfloor \log_p t \rfloor + 1)\rfloor + 1) + \ell}$

5.3 Construction Based on Two p-ary Prefix Codes

For the proposed scheme over $\mathbb{F}_p[x]$, the t-th participant's share $Z_t^{(s)}$ can be regarded as a polynomial of x over $\mathbb{F}_p$. Therefore, we denote by $D(Z_t^{(s)})$ the length of the corresponding p-ary codeword of $Z_t^{(s)}$, then we have

$$D(Z_t^{(s)}) = (k-1)(\ell_t - 1) + \ell, \tag{75}$$

where ℓ_t is the codeword length of using p-ary prefix code to encode positive integer t. If choosing different p-ary prefix codes, the corresponding $D(Z_t^{(s)})$ may be different.

Now, we introduce two p-ary prefix codes named M_1 code and M_2 code. M_1 code is represented by

$$M_1(t) = \overbrace{0, 0, \cdots, 0}^{\lfloor \log_p t \rfloor}[t]_p,$$

where $[t]_p$ is the p-ary expression. Thus, we have

$$L(M_1(t)) = 2\lfloor \log_p t \rfloor + 1. \tag{76}$$

Hence, if using M_1 code as the prefix code, we have

$$D(Z_t^{(s)}) = 2(k-1)\lfloor \log_p t \rfloor + \ell. \tag{77}$$

M_2 code is given by

$$M_2(t) = M_1(\lfloor \log_p t \rfloor + 1)[t]_p.$$

Then the codeword length of $M_2(t)$ is given by

$$L(M_2(t)) = \lfloor \log_p t \rfloor + 2\lfloor \log_p(\lfloor \log_p t \rfloor + 1)\rfloor + 2. \tag{78}$$

Therefore, if using M_2 code as the p-ary prefix code, we have

$$D(Z_t^{(s)}) = (k-1)(\lfloor \log_p t \rfloor) + 2(k-1)(\lfloor \log_p(\lfloor \log_p t \rfloor + 1)\rfloor) + (k-1) + \ell. \tag{79}$$

We tabulate the currently known evolving threshold secret sharing schemes, which are based on p-ary prefix codes and compare the corresponding $D(Z_t^{(s)})$. For the proposed scheme, we analyze the corresponding $D(Z_t^{(s)})$ for using M_2 code as the p-ary prefix code. In Table 2, we also represent the value of the lowest $D(Z_t^{(s)})$ in bold for each case of k. When $k = 2$, the proposed scheme's $D(Z_t^{(s)})$ is lower than the scheme in [30]. When $k \geq 3$, there are no other schemes based on p-ary prefix codes. However, the proposed construction is applicable to any k.

6 Conclusion and Discussion

In this paper, we propose a number of evolving threshold secret sharing schemes. Based on the binary prefix codes, we first propose a construction of evolving 2-threshold scheme for an ℓ-bit secret over on a polynomial ring $\mathbb{F}_2[x]$. Then we extend the construction to evolving k-threshold scheme for $k \geq 3$. Finally, we gave a construction of an evolving k-threshold scheme over a polynomial ring $\mathbb{F}_p[x]$ considering the secret $s \in \{0, 1, \cdots, p-1\}^\ell$. In addition, we show that the share size of the t-th share is $(k-1)(\ell_t - 1) + \ell$, where ℓ_t denotes the codeword length of the integer t encoded by the given binary prefix code. When δ code is applied, the t-th share size is $(k-1)\lfloor \lg t \rfloor + 2(k-1)\lfloor \lg(\lfloor \lg t \rfloor + 1) \rfloor + \ell$, that is smaller than Komargodski's scheme, when $k > 3$ and $\ell > 1$. Third, we consider an ℓ-bit secret $s \in \{0, 1, \cdots, p-1\}^\ell$ for the first time. Based on the p-ary prefix codes, we extend the construction of the evolving k-threshold scheme over a polynomial ring $\mathbb{F}_p[x]$. We also show some p-ary prefix codes for positive integers and then apply the proposed construction to these codes.

There still exists some thought-provoking and challenging issues that remain unresolved, which are also our future works.

1) The correctness and security of the proposed schemes are perfect. If relaxing correctness or security, can one propose more interesting and efficient schemes to achieve high work efficiency or lower share size?

2) Compared with evolving 3-threshold scheme given in [10], the share size of the proposed scheme is larger than the result of [10]. Check whether the proposed scheme be improved to achieve a smaller share size. If not, check whether the scheme [10] be extended for other cases.

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